

DENNIS K. MURIITHI



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SUMMARY

Dennis K. Muriithi is an Associate Professor, Statistician and Lead Data Scientist at the Center for Data Analytics & Modeling in the Faculty of Science and Technology, Chuka University-Kenya. He holds a PhD (Biostatistics) and MSc (Applied statistics) from Moi University and JKUAT-Kenya respectively. He is a dedicated educator and researcher, recognized as an expert in the field of Statistics. His commitment lies in advancing scientific knowledge, mentoring future scientists, and promoting interdisciplinary research. Prof. Muriithi work primarily focuses on Applied Statistics, Data Analytics, Machine Learning and AI Integration with applications in Healthcare, Agriculture, Education and Social sciences. He collaborates with both academia and industry partners to tackle real-world challenges using statistical tools. Prof. Muriithi has published many articles in internationally refereed Journals, recipient of several research grants awards and supervised many PhD and MSc scholars to completion and graduation. Prof. Muriithi is a statistical modeler, data euthanistic, Certified Public Accountant and member of Global Applied Malaria Modeling Network (AMMNet) and International Biometric Society (IBS).

ACCOMPLISHMENTS/ACHIVEMENTS

- **Co-Principal Investigator:** European Union under Global Health EDCTP3 Grant of € **1,299,926** (Euros). One Collaborative Training Program for Outbreak Preparedness and Universal Surveillance (OCTOPUS) project (2026-2028).
- **MalariaAI Project:** Advancing Early Diabetes Detection with Explainable AI
- **Diabetes AI Project:** Advancing Early Diabetes Detection with Explainable AI
- **CardioPredict AI Project:** Predictive Modelling with Explainable AI for Heart Disease Risk Assessment
- **TOP 8% BEST RESEARCHER** in Chuka University according to AD Scientific Index. ID: **4493568** & **Scholar.google.com**
- **AWARD OF FULL SCHOLARSHIP:** Three (3) PhD and Two (2) MSc scholars): Sponsorship for long term training-Food Systems Resilience Project (2023-2029) under my Support and guidance.
- **LEAD APPLICANT AMMNET GRANT AWARD:** Totaling to **\$6,000** USD. <https://ammnet.org/funded-local-event-awards/>
- **LEAD TRAINER/FACILITATOR:** AMMNET Hands-On Workshop on Malaria Modeling held on 7th-8th August 2025 at CDAM-Lab Chuka University: [Github.com](https://github.com)
- **COORDINATOR AND LEAD DATA SCIENTIST:** Center for Data Analytics & Modeling-Chuka University (2025): [CDAM website](https://cdam.chuka.ac.ke)

WEBSITE/PORTFOLIO



SKILLS

- Statistical Analysis using SPSS, Python, R & Rstudio
- Data Visualization using Power BI, SQL, Tableau, R, Python
- Dashboard Design and Development Process using Business intelligent tools
- Machine Learning and Deep Learning
- Data Analytics Expert and Consultant
- Multidisciplinary research
- Thesis supervision
- Generative AI Modeling

- **FACILITATOR:** 2025 Annual AMMNET meeting/Training in Dakar, Senegal. **Titled:** Introduction to R programming & Data Visualization. [Github.com](#) : [AMMnet website](#)
- **FACILITATOR:** Project AIED – UoEm lecture series 2025 **Titled:** Qualtrics: Statistical Data Analysis and AI, Data Export in Qualtrics and Analysis using AI Tools delivered on 11th & 18th March 2025
- **AWARD OF FULL SCHOLARSHIP:** One (1) PhD Sponsorship for long term training - Kenya Climate Smart Agriculture Project (KCSAP) (2019~2023) under my Support and guidance.
- **LEAD TRAINER/FACILITATOR:** AMMNET training Workshop on Malaria Modeling held on 29th-30th August 2024 at Chuka University: [Github.com](#)
- **PUBLICATIONS:** Fifty-Four (54) publications in internationally refereed Journals: [Google Scholar](#)
- **POST GRADUATE SUPERVISION:** 8 PhD and 27 MSc scholars to completion and graduation
- **ATTENDANCE AND CONTRIBUTION:** Fourteen (14) international conferences/workshops/seminars in Kenya, South Africa, Rwanda and Senegal

EXPERIENCE

International Linkages

- **ACU Ambassador**-The Association of Commonwealth Universities, Chuka University ([ACU Ambassador](#))
- **Member**- International Biometric Society, Washington DC & Kenya chapter,
- **Member**- Applied Malaria Modeling Network (AMMnet): USA

Appointment | Committee Involvement

- **Member**- Chuka University ODeL Policy Review Committee
- **Member**- Chuka University human Resource Instruments Customization Committee
- **Member**- Chuka University adhoc Human Resource Advisory Committee
- **Member**- Chuka University establishment of the Directorate of ICT & AI Center Committee
- **Member**- Chuka University staff Establishment Committee
- **Member**- Chuka University Survey Committee

Academic Role

- **Coordinator & Lead Data Scientist**- CDAM | Chuka University
- **Coordinator** -PhD (Applied Statistics), Chuka University
- **Academic leader**- BSc (Applied statistics), Garissa University | Garissa, Kenya | September 2020 - September 2023

Associate Professor of Applied Statistics

Chuka University | Chuka, Kenya | October 2023 - Current

- Teaching and evaluation of undergraduate and postgraduate students.
- Setting, moderating, administering and marking examinations.
- Supervising undergraduate and postgraduate students in research activities.

EDUCATION

Moi University
Eldoret, Kenya
Doctor of Philosophy: PhD.
(Biostatistics)
2018

JKUAT
Nairobi, Kenya
Master of Science: MSc.
(Applied Statistics)
2011

KASNEB
Nairobi, Kenya
Certified Public Accountant:
CPA
2009

Egerton University
Nakuru, Kenya
Bachelor of Science:
(Honours)
2007

Board Member

Directorate of Quality
Assurance, Chuka
University | October 2020 -
October 2023

Patron

Economics and statistics
Club, Chuka University |
January 2013 - January
2019

Field Operations Officer

Ministry of Planning |
Nairobi, Kenya | December
2009 - November 2012

Ass. Monitoring & Evaluation Officer

Ministry of Planning, Kenya
| December 2009 - June
2011

- Developing research proposals for funding.
- Conducting research in the area of my specialization.
- Attending and participating in departmental and faculty meetings and other university committee meetings

Senior Lecturer (Applied Statistics/Statistics)

Chuka University | Chuka, Kenya | June 2019 - October 2023

Lecturer (Statistics)

Chuka University | Chuka, Kenya | June 2016 - June 2019

Research Survey Committee Member

Chuka University | Chuka, Kenya | January 2013 - 2025

To conduct the following surveys;

- Customer Satisfaction Survey, Chuka University
- Staff Competence/Needs Assessment Survey, Chuka University
- Employees Satisfaction Survey, Chuka University
- Work Environment Survey, Chuka University
- Corruption index survey
- HIV Plan Baseline survey
- Effectiveness of Corruption Risk Prevention Strategies, Chuka University
- Service Delivery Improvement Survey, Chuka University

Data Analytics Expert and Consultants

Freelance | September 2009 - Current

- Data science, Machine Learning and AI integration
- Developed excellent working knowledge of industry trends and improvements in processes.
- Customized customer experiences to build brand loyalty.

RESEARCH AND TRAVEL GRANTS

- **Co-Principal Investigator:** European Union under Global Health EDCTP3 Grant of € **1,299,926** (Euros). One Collaborative Training Program for Outbreak Preparedness and Universal Surveillance (OCTOPUS) project (2026-2028).
- **Full Scholarship Award:** Sponsorship for long term training (3 PhD & 2 MSc scholars under my guidance and support) - Food Systems Resilience Project (**2023-2029**).
- **AMMnet Grant** (Lead Applicant): Malaria Risk Prediction in Kenya using Artificial Intelligence and ML Models in R (**2025**) \$ **3,000** (USD). [AMMnet Website](#)
- **AMMnet Travel Grant Award:** Annual AMMnet meeting /training held in Dakar, Senegal, June 25-27 as a **trainer**, Basics: Introduction to R and data visualization.
- **AMMnet Grant** (Lead Applicant): Training Workshop on Malaria

OTHER EXPERIENCES

Internal Examiner (PhD & MSc Thesis)

Chuka University | Chuka, Kenya | January 2018 – Current|

- Conducted thorough evaluation of students' dissertations to maintain high academic standards.
- Reviewed and assessed coursework, ensuring alignment with university grading policies.

Peer Reviewer Member

American Journal of Theoretical & Applied Stats, | 2016 – 2018 |

- Conducted in-depth reviews and analyses of articles for publication.

REFEREES

- **Prof. Mark Okong'o**

Associate professor and Head of Mathematics Section, Department of Physical Sciences, Faculty of Science and Technology, Chuka University, P. o. Box 109 –60400 Chuka~Kenya, Email: markokongo@yahoo.co m, Tel. +254 726322466

- **Dr. Muchiri. J. Githiria,**

Senior Lecturer, School of Mining Engineering, University of the Witwatersrand, Private Bag 3 WITS 2050 South Africa. Email: j.muchirio7@gmail.com

Modeling Software's to Develop Practical Skills in Combating Malaria within the Kenyan Context (2024) \$ 3,000: [AMMnet Website](#)

- **CU Support Project Grant** (Founder): Center for data Analytics & Modeling (CDAM) for Short course training, Consultancy services and Research Hub (2024) Kshs. 5,450,000.
- **Full Scholarship Award:** Sponsorship for long term training (1 PhD Scholar under my guidance and support to graduation 2024) - Kenya Climate Smart Agriculture Project (KCSAP) (2019~2023)
- **CU internal Research Grant** (Principal Investigator): A study on Optimizing of cucumber growth parameters and Yield to Organic Manure Using Response Surface Methodology and Central Composite Design (2022~2024). KShs 700,000.
- **TU internal Research Grant** (Co-Principal Investigator): A study on Optimization of Biodiesel and Glycerol Production from Mango Seed Oil using Response Surface Methodology and Central Composite Design (2023~2025) KShs 200,000.
- **CU internal Research Grant** (Principal Investigator): A study on Modeling and Optimization of Multiple Responses of Watermelon to Organic Manure Using Response Surface Methodology. (2016~2017)". KShs 200,000.

CONFERENCE/WORKSHOP ATTENDANCE & CONTRIBUTIONS

- **Short Course on Research Skills Development**
Organized under the Research Community Training Grant, hosted by the ACU Supporting Research Community and the University of Port Harcourt, Nigeria and Sponsored by the Association of Commonwealth Universities (ACU), United Kingdom (29th Sep-3rd Oct 2025)
- **Trainer/Facilitator 2025 Annual AMMnet meeting/Training in Dakar, Senegal.** Topic: Introduction to R programming & Data Visualization. (25th-27th June 2025)
- **NAMBARI (Moi Brown Partnership For Biostatistics Training) Workshop:** *Advanced Bio-statistical Methods: survival Analysis and Machine Learning in health Sciences (14th ~17th July 2025)*
- **10th Chuka University International Research Conference: Presentation:** *Application of response surface methodology for Optimization of cucumber (cucumis sativus L) yield Parameters using second order rotatable design (5th ~ 6th October, 2023)*

SELECTED PUBLICATIONS

- Lumumba, V.W, Muriithi. D.K, Monicah Oundo (2026). A Transparent AI-Driven Ensemble Learning Approach for Predicting Mental Health Risks among University Students in Kenya, *IJPBS*, 11(2): pp 35-51
- Waburi, E. W., Muriithi, D. K., & Sundays, E. M. (2026). Early Detection of Hypertension Using Explainable AI Model: Random Forest Model. *European Journal of Artificial Intelligence and Machine Learning*, 5(1), 1-6.
- Monicah N., Kaimenyi, C., Muriithi.D.K, Thomas Gordon & Deborah A. Gust(2025). Factors associated with psychological distress in mothers and girls in Kenya: A baseline analysis. *ASFI Research Journal*, 2(1), 1-19
- Waburi, E. W., Muriithi, D. K., & Sundays, E. M. (2025). Integrating explainable machine learning models for early detection of hypertension: A transparent approach to AI-driven healthcare. *American Journal of Artificial Intelligence*, 9(2), 154-166
- Lumumba, V. W., Muriithi, D. K., Njoroge, E. W., Langat, A. K., Mwebesa, E., & Wanyama, M. A. (2025). An explainable AI framework for neonatal

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• Dr.Vincent Michael Kiberu

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Tel: (+256) 701 431524

mortality risk prediction in Kenya: Enhancing clinical decisions with machine learning. *Biomedical Statistics and Informatics*, 10(3), 64-83.

- Lumumba, V.W, **Muriithi. D.K**, Monicah Oundo (2025). Evaluating the Performance of Selected Single Classifiers with Incorporated Explainable Artificial Intelligence (XAI) in the Prediction of Mental Health Distress Among University Students, *AJAI*, 9(2): pp 133-144
- Gitonga, C. K., **Murithi, D.**, & Chebet, E. (2025). Mitigating demographic bias in ImageNet: A comprehensive analysis of disparities and fairness in deep learning models. *European Journal of Artificial Intelligence and Machine Learning*, 4(2), 15–26
- **Muriithi, D. K.**, Lumumba, V. W., Olawale, O. A., & Muriith, D. M. (2025). Explainable artificial intelligence models for predicting malaria risk in Kenya. *European Journal of Artificial Intelligence and Machine Learning*, 4(1), 1–8
- Waka O, **Muriithi.D.K** and Okongo. M.O. (2025). Statistical Modelling of Staff Survival Time in Service at Chuka University. *AJPS* 27 (4):17-27.
- Remmy J. N, **Muriithi, D.K** & Onesmus Mbaabu. (2024). Effect of Public Wages on External Debt in Kenya. *Science Research*, 12(5), 109-116.
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Number of publications (54) in internationally refereed Journals